

FEDLEY SIKOLIA

Electronic & Computer Engineer

Expected Graduation: August 2026 · JKUAT

📍 Nairobi, Kenya

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PROFILE SUMMARY

Final-year Electronic & Computer Engineer (Expected: August 2026) with hands-on experience across vulnerability lifecycle analysis, embedded IoT systems, and full-stack backend development. Bridges hardware and software end-to-end — from ESP32 firmware and M-Pesa payment integration to Django REST APIs, phishing simulation platforms, and SIEM-based detection engineering. Delivered 8 production-grade projects including an authorized penetration test achieving root shell access on a simulated fintech environment, and a live IoT access control system processing real M-Pesa transactions. Builds production-ready, security-conscious systems with measurable real-world impact.

CORE SKILLS

Security & Analysis	Nmap · Metasploit · Searchsploit · Wireshark · Splunk · CVE Analysis · PCAP Forensics · UFW · Fail2ban
Programming	Python · JavaScript · C++ · SQL · Embedded C
Backend & Web	Django · REST APIs · React · Node.js · Express · JWT Auth · MongoDB · MySQL · HTML/CSS
Embedded & IoT	ESP32 · Arduino · SIM800L GSM · KiCad · Proteus · PCB Design · Relay Logic · Sensors
Cloud & AI	Azure · Google Cloud · Docker · Kali Linux · Prompt Engineering · LLM Applications
Tools & Workflow	Git · GitHub · Agile · Trello · Technical Documentation

EXPERIENCE

Security Research & Vulnerability Analysis · JKUAT — *EEC 2505 Cryptography & Network Security* Feb – May 2026

NexaFinance Breach Simulation · Authorized Lab Environment

- Conducted end-to-end penetration test in a Dockerized environment — enumerated 20+ exposed services with Nmap, mapped 6 CVEs via Searchsploit, and achieved root shell access via Metasploit (vsftpd 2.3.4 + Samba exploits).
- Extracted and cracked /etc/shadow credentials using John the Ripper; reconstructed full attack chain forensically via Wireshark PCAP, extracting exploit trigger timestamps and IOC signatures.
- Built Splunk SIEM detection queries and custom alerts; validated full hardening (UFW firewall rules, Fail2ban, service disabling) — reduced attack surface from **20+ open ports to under 5**, with re-run exploits confirming closure.

Software Development Attaché · JHUB Africa

June – August 2025

- Led backend development of SecureNet AI — a full-stack phishing simulation platform supporting **3 user roles (superadmin / admin / analyst)** and logging **140+ simulated phishing events** with real-time analytics.
- Implemented JWT-secured authentication, structured phishing event logging, and CSV export pipelines using Django and React across a 3-month agile delivery cycle.
- Designed dashboards visualizing campaign performance metrics, click-through rates, and simulated credential submission patterns for security team analysis.

Technical Assistant Attaché · Fedha Electrics

May – August 2024

- Designed and delivered responsive websites for multiple business clients; developed PCB schematics using KiCad and Proteus for electromechanical projects.

Operations & Maintenance Attaché · *Kenya Power & Lighting Company (KPLC)*

June – August 2023

- Supported LV/HV distribution infrastructure maintenance, field diagnostics, and safety compliance checks across live grid environments.

KEY PROJECTS

SecureNet AI — Cybersecurity Platform · *June – Aug 2025* · github.com/sikolia-2/securanet-ai-website · github.com/sikolia-2/SecureNet-phishing

Ethical phishing simulation platform with JWT RBAC, SHA-256 credential masking, 140+ event analytics, campaign lifecycle management, and CSV export. Stack: Node.js · Express · MongoDB · React · Django.

Automated Door Lock System with M-Pesa · *Jan 2026 – Present* · github.com/sikolia-2/door-lock-system

Production-ready IoT access control for self-service storage. ESP32 triggers M-Pesa STK push (Daraja API), bills KES 2/min, with 4 access modes, SIM800L GSM backup, tamper detection, and Django REST backend (8 endpoints). Stack: ESP32 · Django · Daraja API · SIM800L.

Electricity Metering & Theft Detection · *Dec 2025 – Present* · github.com/sikolia-2/Blockchain-Metering-System

Embedded monitoring system using current sensors and anomaly-detection logic to flag irregular consumption patterns indicative of tampering, with blockchain-based audit logging. Stack: Arduino · Current Sensors · Python · SQL.

SipSwift — Cross-Platform Food Delivery App · *2026* · github.com/sikolia-2/Food-Delivery-App

Flutter delivery app: OTP auth, Google Sign-In, M-Pesa checkout, live order tracking, and dedicated rider dashboard across 9 screens and 3 service layers. Stack: Flutter · Dart · REST API.

Smart Waste Management System · *Jan – Mar 2024* · github.com/sikolia-2/Smart-Waste-Management-System-Project

IoT waste collection system with ultrasonic fill-level sensors, dynamic route optimization, and automated scheduling. Led hardware design and embedded firmware. Stack: IoT · Embedded Firmware · Python · SQL.

EDUCATION

BSc. Electronic & Computer Engineering · *Jomo Kenyatta University of Agriculture and Technology (JKUAT)* *Sept 2021 – Aug 2026*

- Relevant coursework: Cryptography & Network Security · Embedded Systems · Digital Electronics · Control Theory · Software Engineering.

CERTIFICATIONS

- **Endpoint Security** — Cisco Networking Academy · May 2026 · ID: bc8caecd-2808-4f0e-b6f9-3d5073f0a2a4
- **AI for Sustainable Development: Strategies & Models of Tech-Based Innovations** — ITACA Project / Università Cattolica del Sacro Cuore · April 2026
- **Building LLM Applications with Prompt Engineering** — NVIDIA · ID: TxBCSuwYTgOd-zwFULwpoA
- **Cybersecurity Essentials** — Atingi · EU4Digital Facility
- **Machine Learning: Data Preparation & Feature Engineering** — Atingi
- **Virtual Assistant Program** — ALX Africa · ID: s62r39F7nh

AFFILIATIONS

- ALX Africa – AI Career Essentials (AiCE) Program: Practical experience with AI tools, prompt engineering, and applied web development projects.
- Society of Engineering Students (SES), JKUAT: Engineering discourse, technical workshops, and peer-led innovation initiatives.

REFEREES

Available upon request.